

Nicolas Romanella

Mechanical Engineer | CAE, Multibody & Structural Dynamics

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Mechanical engineer specialized in computer-aided engineering, with experience in multibody dynamics, structural dynamics, control theory, model-based development, and simulation-driven engineering. Focused on efficient workflows, process automation, and data analysis, with international experience at Volvo Cars and leadership experience in a motorsport engineering team.

ENGINEERING EXPERIENCE

Vehicle Dynamics Engineer

Göteborg, Sweden

Volvo Cars

Jan 2026 – Present

Working on the integration of reduced flexible Body-in-White (BIW) and structural component models into real-time vehicle dynamics simulations. Analyzing the impact of structural flexibility on ride comfort, handling, and steering response in passenger cars.

Head of Simulation & Modeling

Modena, Italy

Impulse Modena Racing

Nov 2024 – Dec 2025

Promoted to lead a five-member team within the Vehicle Dynamics Division, focusing on improving model accuracy. Trained new members in simulation tools, supported multiple on-track validation tests across major Italian circuits. Selected to compete in the MotoStudent International Competition at MotorLand Aragón.

Vehicle Dynamics Engineer

Modena, Italy

Impulse Modena Racing

Nov 2023 – Oct 2024

Developed and validated motorcycle dynamic models used to predict on-track performance and assist design decisions. Built software tools for telemetry analysis and GPS-based data processing during testing. Gained hands-on experience supporting mechanical setup and data collection during test sessions.

ADDITIONAL EXPERIENCE

Business Support Assistant

Benevento, Italy

Nuova Coperbit SRLS

Nov 2022 – Dec 2025

Provided financial and strategic support to a construction materials company. Prepared estimates, invoices, and investment projections to assist in decision-making. Developed tax and cash-flow forecasts and managed the sale and nationwide shipping of unused equipment.

Student Representative

Pisa, Italy

University of Pisa

Dec 2019 – Sep 2022

Elected to represent students living in university residences, acting as the main point of contact between students and DSU Toscana. Led meetings to improve residence services, coordinated relocations, and organized community events to foster collaboration and mutual respect.

EDUCATION

Master of Science in Automotive Engineering

Modena, Italy

University of Modena and Reggio Emilia

Sep 2023 – Jun 2026

GPA: 3.91/4.00.

Summer School in Nuclear Engineering

Champaign, IL

University of Illinois Urbana-Champaign

Aug 2023 – Sep 2023

Grade: 4.00/4.00.

Bachelor of Science in Mechanical Engineering

Pisa, Italy

University of Pisa

Sep 2018 – Aug 2023

GPA: 3.75/4.00.

SOFTWARE & TOOLS

CAE MSC Adams, MSC Nastran, VI-Grade, Adams Car, ANSA

CAD Siemens NX, CATIA V5/V6, SolidWorks

Programming MATLAB, Simulink, Python, C++